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Cashless Payment and Financial Inclusion*

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*Paper link: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3948925

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1 Executive overview

1.1 What is the paper asking?

The paper asks a clean causal question:

Does adopting cashless payment (mobile payment) increase consumer credit access and credit quantity, especially for underserved populations?

The key economic mechanism is that a BigTech platform (Alipay) can observe *transaction-level payment flows* and use them to screen and scale consumer credit.

1.2 Why is this hard empirically?

Payment behavior and credit are jointly determined:

- consumers with higher income/financial literacy may both use cashless payment more and qualify for more credit;
- the platform may deliberately expand payment and credit together (bundling/synergies), creating simultaneity;
- credit access itself can increase payment flow (relaxed liquidity constraint), so causality runs both ways.

Hence, OLS regressions of credit on payment flow are likely biased.

1.3 Core identification strategy (instrumental variables)

The paper exploits a natural experiment:

the staggered rollout of Alipay-bundled dockless shared bikes across Chinese cities.

When shared bikes arrive, users often must use the Alipay wallet to unlock/pay via QR code, which nudges them into using Alipay for *other* in-person transactions. City-month shared-bike *placement intensity* becomes an instrument for individual-level in-person payment flow.

1.4 Main results (headline magnitudes)

Using monthly panel data on 41,485 Alipay users (May 2017–Sep 2020) and 2SLS:

- Payment flow causally increases credit access.** A month with positive in-person payment flow increases the probability of having access to Alipay’s virtual credit card by about **56%** (linear-probability interpretation).
- Payment flow causally increases credit limits.** Conditional on access, a month with positive in-person payment flow raises the (log) credit line by about **2.03** (roughly **+203%** in levels). A 1% increase in in-person payment flow increases the credit line by about **0.41%**.
- Effects are stronger for underserved groups** (less educated, older), consistent with a *financial inclusion* mechanism.

1.5 Interpretation

Cashless payment adoption generates transaction data that improves screening and expands BigTech consumer credit supply, with disproportionately large benefits for users who previously lacked traditional financial access.

2 Institutional background and why the setting is unusually clean

2.1 Alipay + Huabei: “credit supply” can be observed

A distinctive feature of Alipay’s flagship consumer credit product (Huabei) is that:

- users do not go through a standard application process to learn eligibility and an approximate credit line,
- the platform reveals access and the credit line essentially automatically.

This matters because it reduces the classic problem in consumer credit studies: observed borrowing confounds credit demand and credit supply. Here, *credit access* and *credit line* are closer to supply-side objects.

2.2 Why shared bikes generate plausibly exogenous variation in payment adoption

Dockless bikes create a recurring need for QR-code payments. When bikes are bundled with Alipay, the bike market generates an external “technology adoption push” for the wallet:

- placement of bikes changes the local availability of the bike service,
- using the bikes requires interacting with (and setting up) the payment interface,
- users then substitute into Alipay for other in-person payments (a behavioral nudge).

The identifying premise is that *conditional on fixed effects*, bike placement shocks are driven by bike-industry competition and operational constraints, not by time-varying unobserved factors that directly change Alipay credit decisions for local residents.

3 Data and key variables

3.1 Main sample

The empirical analysis uses proprietary monthly panel data for:

- 41,485 randomly sampled Alipay users,
- May 2017–September 2020,
- linked demographics + detailed payment categories + bike usage + credit access/limits + assets under management (AUM).

3.2 Key outcomes: credit provision

- **Credit access** $CreditAccess_{i,t} \in \{0, 1\}$: whether user i has access to Huabei at month t .
- **Credit line** $CreditLine_{i,t} > 0$: the limit (conditional on access).

3.3 Key endogenous regressor: in-person payment flow

Let $IPF_{i,t}$ denote in-person Alipay payment flow in month t . The paper uses different transforms $g(\cdot)$ to address zeros and skewness:

$$g(IPF_{i,t}) \in \{\log(1 + IPF_{i,t}), \mathbf{1}\{IPF_{i,t} > 0\}, \log(IPF_{i,t}) \cdot \mathbf{1}\{IPF_{i,t} > 0\}\}.$$

3.4 Instrument: city-month bike placement

Let $BP_{c,t}$ be the number of active Alipay-bundled shared bikes placed in city c in month t . The baseline instrument is:

$$\log(BP_{c,t}).$$

4 All models in the paper

4.1 Main 2SLS system (core empirical design)

First stage (payment flow on bike placement).

$$g(IPF)_{i,t} = a_1 + b_1 \log(BP)_{c,t} + d_i + \tau_t + \varepsilon_{i,t}^{(1)}. \quad (1)$$

Second stage (credit provision on instrumented payment flow).

$$Y_{i,t} = a_2 + b_2 \widehat{g(IPF)}_{i,t} + d_i + \tau_t + \varepsilon_{i,t}^{(2)}, \quad (2)$$

where $Y_{i,t}$ is either $CreditAccess_{i,t}$ or $\log(CreditLine_{i,t})$.

OLS benchmark (endogenous).

$$Y_{i,t} = a_0 + b_0 g(IPF)_{i,t} + d_i + \tau_t + u_{i,t}. \quad (3)$$

Fixed effects and inference. All baseline specifications include **individual fixed effects** d_i and **year-month fixed effects** τ_t . Standard errors are clustered (two-way) by city and year-month in the key panel specifications.

4.2 Event-study specifications (Figures 4, 5, A.4, A.8)

Two event-study regressions clarify the mechanism and timing.

(A) User-level adoption event study: payment behavior around first bike usage. Let event time k be months relative to user i 's first bike usage month. A typical specification is:

$$y_{i,t} = \alpha + \sum_{k=-5}^4 \beta_k \mathbf{1}\{EventTime_{i,t} = k\} + \beta_{5+} \mathbf{1}\{EventTime_{i,t} \geq 5\} + d_i + \tau_t + \epsilon_{i,t}, \quad (1)$$

where $y_{i,t}$ is either $\log(1 + \text{In-person non-bike payment})$ or $\mathbf{1}\{\text{positive non-bike payment}\}$. The omitted category is $k = -1$.

(B) City-level placement event study: staggered placement shocks. Let event time k be months relative to city c 's largest bike-placement shock month. A typical specification is:

$$\widetilde{BP}_{c,t} = \alpha + \sum_{k=-5}^4 \beta_k \mathbf{1}\{EventTime_{c,t} = k\} + \beta_{5+} \mathbf{1}\{EventTime_{c,t} \geq 5\} + \delta_c + \tau_t + \nu_{c,t}, \quad (2)$$

where $\widetilde{BP}_{c,t}$ is normalized bike placement in $[0, 1]$ and δ_c are city fixed effects.

4.3 Econometric framework in the Online Appendix (simultaneity and OLS bias)

The Online Appendix formalizes the two-way relationship between payment and credit:

Credit provision (BigTech sets credit as a function of payment).

$$CL_{i,t} = a_0 + a_1 IPF_{i,t} + d_i + \tau_t + \eta_{i,t}^{OV} + \eta_{i,t}^{EE}. \quad (A1)$$

Payment flow decision (consumer uses payment more when credit is higher).

$$IPF_{i,t} = b_0 + b_1 CL_{i,t} + \mu_i + \omega_t + \zeta_{i,t}. \quad (A2)$$

IV identification. If bike placement $BP_{c,t}$ is exogenous and relevant, IV identifies a_1 :

$$\widehat{a}_1^{IV} = \frac{\text{Cov}(CL_{i,t}, BP_{c,t})}{\text{Cov}(IPF_{i,t}, BP_{c,t})} = a_1. \quad (A3)$$

4.4 Illustrative theory in the Online Appendix (Figure 6)

The appendix also gives a simple lender problem to illustrate how better information changes credit limits.

Objects. A continuum of borrowers has type $q_i \sim U[0, 1]$ (creditworthiness). The lender chooses a credit limit $l_i \geq 0$.

Key idea. A “cashless-payment shock” improves the lender’s information about q_i (e.g., through transaction-level signals), which changes the optimal lending rule and can increase inclusion for some types while tightening screening for others. Figure 6 visualizes this logic.

5 Identification: why bike placement is a credible instrument

5.1 Relevance: bike placement moves payment flow

The relevance condition requires a strong first stage in (1). The paper provides multiple layers of evidence:

- **Direct first-stage regressions:** bike placement predicts in-person payment flow (Table 2; Table 4 first-stage panel).
- **Behavioral-nudge evidence:** after a user tries shared bikes, non-bike in-person payments jump sharply and persist (Figure 4; Figure A.4).
- **Intensity/monotonicity:** effects are stronger for heavier bike users (Table 3; Figure A.5; Table A.5).

5.2 Exclusion restriction: bike placement affects credit only through payment

The identifying assumption is:

$$\mathbb{E}\left[\varepsilon_{i,t}^{(2)} \mid \log(BP)_{c,t}, d_i, \tau_t\right] = 0,$$

i.e., conditional on fixed effects, bike placement is not correlated with unobserved shocks to credit access/limits.

The paper addresses three main threats:

Threat 1: common city-time shocks. Mitigated by fixed effects and city-level checks that show no robust correlation between bike placement and macro conditions (Table A.4), and by specifications that absorb city×year-month variation (Table A.9).

Threat 2: bike usage has a direct effect on credit (not via payment). Addressed via designs that separate bike usage from payment flow using two-IV strategies (Tables A.10–A.11), plus checks showing bike usage loses explanatory power once payment is controlled for in stable-payment subsamples (Table A.12).

Threat 3: predictable rollout / bunching. Placement timing is staggered across cities (Figure 5; Figure A.6), supporting plausibly exogenous shocks.

6 Main empirical findings: payment flow $\rightarrow$ credit provision

6.1 Core causal estimate: 2SLS vs OLS (Table 4)

Table 4 is the central result. It estimates b_2 in (2) for both margins and multiple payment transforms.

Extensive margin (credit access). Instrumented in-person payment flow increases the probability of having Huabei access.

Intensive margin (credit line conditional on access). Instrumented in-person payment flow increases credit limits substantially.

Why OLS is much smaller. OLS is an order of magnitude smaller than IV, consistent with endogeneity and attenuation from simultaneity and omitted variables.

7 Mechanisms: information vs enforcement and consumer behavior

7.1 Information channel: noncredit transactions still predict credit (Table 6)

Separating payment into noncredit vs credit-paid components, the paper shows that *noncredit* payment flow causally increases credit access and limits, supporting the idea that the platform uses spending data as an information signal even before credit history forms.

7.2 Enforcement channel: controlling for AUM (Table 7)

If platform assets serve as implicit collateral, AUM could explain credit decisions. Controlling for AUM does not attenuate the IV effect of payment flow, suggesting enforcement is not the main driver in these regressions.

7.3 Consumer behavior (Table 5 and Table A.17)

Instrumented payment flow increases virtual-credit-card usage shares, but does not increase compulsive spending shares in Table 5. Table A.17 further shows that higher compulsive spending shares are negatively correlated with credit access and credit lines, consistent with platform screening against risky consumption patterns.

8 Financial inclusion: who benefits most?

8.1 Who is underserved? (Table 8)

Less educated and older users have fewer linked bank cards, lower AUM, less investment-service usage, and weaker profile/verification completion, consistent with being underserved by traditional finance.

8.2 Heterogeneous causal effects (Table 9 and Table A.16)

The causal effect of payment flow on credit access and limits is concentrated among the less educated (Table 9) and is stronger for older users (Table A.16). This is the paper’s key *financial inclusion* result.

9 Interpretation of all main figures (Figures 1–6)

9.1 Figure 1: Mobile payment penetration across countries

Purpose. Establish global context: mobile payments vary dramatically across countries.

Interpretation. China is an extreme adopter: high transaction volume and penetration relative to GDP, motivating China as a laboratory for “going cashless.”

9.2 Figure 2: Mobile vs card payment in China and the US

Purpose. Show the macro time-series transition.

Interpretation. China’s mobile-payment volume rises rapidly relative to GDP compared to US card payments, consistent with a large regime change in payment technology.

Figure 1: Mobile Payment Penetration across Countries
These figures show the GDP-adjusted mobile payment transaction volume per user and the mobile payment penetration rate for selected countries in 2019 and 2023. Data sources are the Statista Digital Market Outlook and the World Bank.

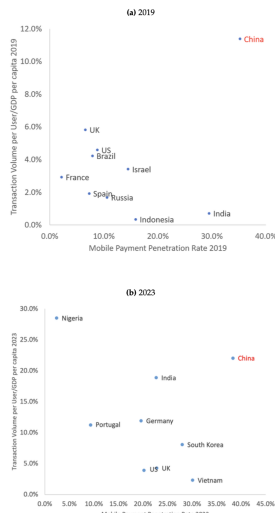


Figure 2: Transaction Volume of Mobile and Card Payment in China and US
This figure presents the time series of the GDP-adjusted transaction volume of mobile and card payments in China and the US from 2012 to 2021. Data sources are the US Federal Reserve, the People’s Bank of China (PBOC), and the World Bank.

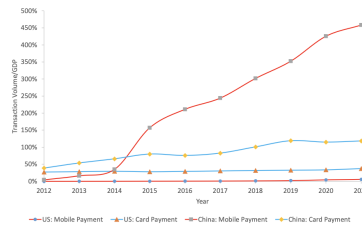
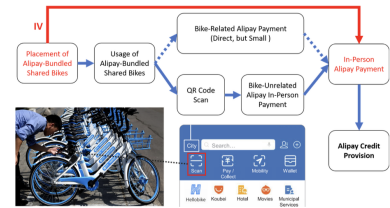


Figure 3: Logic Flow of the Instrumental Variable
This figure presents a graphical illustration of the mechanisms that show how the city-wide placement of Alipay-bundled shared bikes affects the city’s residents’ in-person Alipay payment at the individual level.



9.3 Figure 3: Logic flow of the instrumental variable

Purpose. Clarify the identifying chain.

Interpretation.

Bike placement → Bike adoption/usage → In-person payment flow → Credit access/limits.

It also highlights possible violations (direct bike-to-credit links), motivating robustness tests.

9.4 Figure 4: Bike adoption and non-bike payment flow (user event study)

Purpose. Provide mechanism evidence that bike adoption spills over into broader payment adoption.

Interpretation. Non-bike in-person payments jump at first bike usage and persist, supporting a behavioral nudge story (setup costs, learning) rather than reverse causality.

Figure 4: Bike Adoption and Non-bike Payment Flow

This figure plots the β_1 coefficients estimated in the following regressions:

$$\text{Panel (a): } \log(1 + \text{In-Person Non-Bike Payment Flow})_{i,t} = a_0 + \sum_{\tau=-5}^5 \beta_1 \cdot \mathbb{1}(t = \tau) \cdot \mathbb{1}(\tau \neq -1) + \beta_2 \cdot \mathbb{1}(t \geq 5) + \delta_i + \mu_t + \epsilon_{i,t},$$

$$\text{Panel (b): } \mathbb{1}(\text{In-Person Non-Bike Payment Flow} > 0)_{i,t} = a_0 + \sum_{\tau=-5}^5 \beta_1 \cdot \mathbb{1}(t = \tau) \cdot \mathbb{1}(\tau \neq -1) + \beta_2 \cdot \mathbb{1}(t \geq 5) + \delta_i + \mu_t + \epsilon_{i,t},$$

where $\log(1 + \text{In-Person Non-Bike Payment Flow})_{i,t}$ is the $\log(1 + x)$ transformed amount of in-person payments on purchases not directly related to the usage of shared bikes by individual i at time t using Alipay, and $\mathbb{1}(\text{In-Person Non-Bike Payment Flow} > 0)_{i,t}$ indicates whether this amount is positive. t corresponds to the number of months after each individual's month of the first usage of Alipay-bundled shared bikes, δ_i is the individual fixed effects, μ_t is the year-month fixed effects, and $\epsilon_{i,t}$ is the error term that varies across individuals and over time. The sample covers only users who used the Alipay-bundled shared bikes at least once in the sample period, which is from May 2017 to September 2020. For each bike user, the sample only covers periods in which the t is not earlier than -5 .

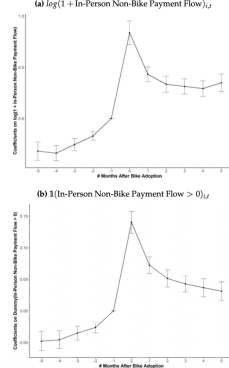


Figure 5: Staggered Placement of Shared Bikes

This figure plots the β_1 coefficients estimated in the following regression:

$$\text{Normalized Bike Placement}_{i,t} = a_0 + \sum_{\tau=-5}^5 \beta_1 \cdot \mathbb{1}(t = \tau) \cdot \mathbb{1}(\tau \neq -1) + \beta_2 \cdot \mathbb{1}(t \geq 5) + \delta_i + \mu_t + \epsilon_{i,t},$$

where Normalized Bike Placement $_{i,t}$ is defined as $\frac{\text{Bike Placement}_{i,t}}{\sum_{i=1}^N \text{Bike Placement}_{i,t}}$, which is a measure with a range of $[0, 1]$. t corresponds to the number of months after each city's month with the largest bike-placement shock, δ_i is the city fixed effects, μ_t is the year-month fixed effects, and $\epsilon_{i,t}$ is the error term that varies across cities and over time. The sample period is from May 2017 to January 2020, which avoids the later COVID lockdown periods. For each city, the sample only covers periods in which the t is not earlier than -5 .

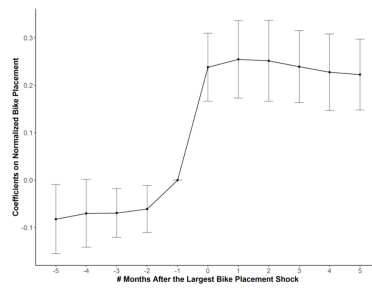
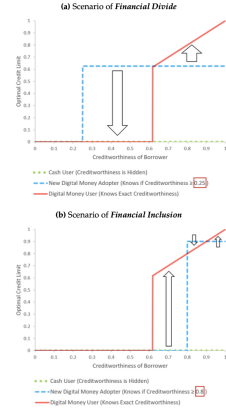


Figure 6: Cashless Payment's Financial Inclusion Implication: An Example

The figures illustrate how a lender sets optimal credit limits for heterogeneous borrowers in an economy with a continuum of borrowers. The illustrative example's setup is in Section A.2. Given knowledge about the borrower's type, the lender chooses the optimal credit limit to maximize the expected profit. There are three cases with different information sets for the lender based on the borrower's digital payment adoption stage. In stage one (green dotted line), borrowers use cash and the lender knows nothing about their creditworthiness. In stage two (blue dash line), borrowers adopt digital money and the lender knows if their creditworthiness is above a threshold. In stage three (red solid line), borrowers use digital money and the lender knows their exact creditworthiness. Figure (a) shows the financial divide scenario, where some less creditworthy borrowers lose in the transition from stage two to three. Figure (b) shows the financial inclusion scenario, where some less creditworthy borrowers gain in the transition to stage three.



9.5 Figure 5: Staggered placement of shared bikes (city event study)

Purpose. Support exogeneity of placement timing.

Interpretation. Bike placement shocks are staggered across cities; peaks occur at different times, inconsistent with a single national shock driving both placement and credit.

9.6 Figure 6: Financial inclusion implication (theoretical example)

Purpose. Provide intuition for how better transaction information affects lending rules across types.

Interpretation. Digital payment improves the lender's information set, expanding credit for some types and potentially sharpening screening at the margin; inclusion gains can be heterogeneous and non-monotonic.

10 Interpretation of all main tables (Tables 1–9)

10.1 Table 1: Summary statistics

Key takeaways.

- Users are active for most months; about 62% have Huabei access in a typical month.
- Bike usage is common but not universal (about 29% ever use bikes), creating meaningful variation.

- Virtual-credit-card shares are sizable; compulsive-spending shares are low on average.

10.2 Table 2: Bike placement effects on payment flow and credit (relevance + reduced form)

Panel A (relevance). Bike placement increases in-person payment flow, especially for bike users and after first bike usage.

Panel B (reduced form). Bike placement increases credit lines; interactions suggest the bike-adoption channel (not generic city shocks).

Table 1: Summary Statistics

The table summarizes the key variables for our analysis of 41,485 Alipay users from May 2017 to September 2020. The variables are categorized into three types at different levels. At the individual level, # Active months, counts the months with payment activities; Is Male, is a dummy for male; Low Education, is a dummy for below bachelor's degree; Birth Year, records the birth year; Bike User, is a dummy for using a shared bike at least once. At the city-month level, $\log(\text{Bike Placement})_{i,t}$ is the log number of active shared bikes in city c at time t . At the individual-month level, Credit Access _{i,t} is a dummy for having access to Alipay's virtual credit card at time t ; $\log(\text{Credit Line})_{i,t}$ is the log credit line conditional on Credit Access _{i,t} ; $\log(\text{In-Person Payment Flow})_{i,t}$ and $\log(\text{Online Payment Flow})_{i,t}$ are the log amounts of in-person and online payments using Alipay; Virtual Credit Card Share in In-Person Payment _{i,t} and Virtual Credit Card Share in Online Payment _{i,t} are the shares of in-person and online payments using the virtual credit card; Compulsive Spending Share in In-Person Payment _{i,t} measures the share of in-person Alipay payments made by individual i at time t for cigarettes, games, and lotteries; Compulsive Spending Share in Online Payment _{i,t} measures the share of online Alipay payments made by individual i at time t for cigarettes, games, lotteries, or live streaming services.

	N	Mean	Std	Min	p25	Median	p75	Max
Individual Level								
# Active months	41,485	31.86	11.38	1.00	24.00	37.00	41.00	61.00
Is Male	41,214	0.54	0.50	0.00	0.00	1.00	1.00	1.00
Low Education	41,489	0.08	0.23	0.00	1.00	1.00	1.00	1.00
Birth Year	41,214	1,983.38	12.75	1,930.00	1,970.00	1,985.00	1,995.00	2,014.00
Bike User	41,485	0.29	0.45	0.00	0.00	0.00	1.00	1.00
City-Month Level								
$\log(\text{Bike Placement})_{i,t}$	12,665	7.08	3.59	0.00	4.11	7.85	9.91	13.91
Individual Month Level								
Credit Access _{i,t}	1,321,837	0.62	0.49	0.00	0.00	1.00	1.00	1.00
$\log(\text{Credit Line})_{i,t}$	819,812	7.88	1.58	3.00	6.91	8.13	9.13	11.02
$\log(\text{In-Person Payment Flow})_{i,t}$	68,428	5.70	2.29	4.41	4.31	6.04	7.27	15.68
$\log(\text{Online Payment Flow})_{i,t}$	843,995	5.76	1.80	4.41	4.70	5.88	6.93	15.74
Virtual Credit Card Share in In-Person Payment _{i,t}	684,428	0.54	0.42	0.00	0.00	0.04	0.82	1.00
Virtual Credit Card Share in Online Payment _{i,t}	843,995	0.53	0.41	0.00	0.00	0.01	0.80	1.00
Compulsive Spending Share in In-Person Payment _{i,t}	684,428	0.03	0.14	0.00	0.00	0.00	0.30	1.00
Compulsive Spending Share in Online Payment _{i,t}	843,995	0.01	0.10	0.00	0.00	0.00	0.30	1.00

Table 2: Effects of Bike Placement on Payment and Credit

These tables report the effects of city-level placement of shared bikes on the individual-level in-person payment flow and digital credit access. $\log(\text{Bike Placement})_{i,t}$ is the log number of active shared bikes in city c at time t . Bike User _{i,t} is a dummy for using a shared bike at least once from May 2017 to September 2020. After First Bike Usage _{i,t} is a dummy for after the first bike usage. $\log(1 + \text{In-Person Payment Flow})_{i,t}$ and $\log(1 + \text{Credit Line})_{i,t}$ are the $\log(1 + x)$ transformed amounts of in-person payment flow and credit line through Alipay in CNY. Panel A shows the effects on payment flow and Panel B shows the effects on credit line. Columns (1) and (2) use individual and year-month fixed effects with the full sample. Column (3) adds city times year-month fixed effects with the bike user sample. All standard errors are clustered at city and year-month level. ***, **, and * denote the 1%, 5%, and 10% confidence level, respectively. I report standard errors in parentheses.

Panel A. Bike Placement and Individual-Level In-Person Payment Flow

	(1)	(2)	(3)
$\log(\text{Bike Placement})_{i,t}$	0.041*** (0.010)	0.011 (0.009)	
Bike User _{i,t} × $\log(\text{Bike Placement})_{i,t}$		0.103*** (0.017)	
After First Bike Usage _{i,t}		-0.123 (0.161)	
After First Bike Usage _{i,t} × $\log(\text{Bike Placement})_{i,t}$		0.049*** (0.014)	
Individual FE	YES	YES	YES
Year-Month FE	YES	YES	-
City × Year-Month FE	NO	NO	YES
Clustered by City and Year-Month	YES	YES	YES
Sample	Full Sample Full Sample Bike Users		
Observations	1,238,309	1,238,309	435,872
Adjusted R ²	0.351	0.552	0.490

Note: ***p < 0.01; **p < 0.05; *p < 0.1

Panel B. Bike Placement and Individual-Level Digital Credit Line

	$\log(1 + \text{Credit Line})_{i,t}$		
	(1)	(2)	(3)
$\log(\text{Bike Placement})_{i,t}$	0.027*** (0.008)	0.009 (0.010)	
Bike User _{i,t} × $\log(\text{Bike Placement})_{i,t}$		0.069** (0.023)	
After First Bike Usage _{i,t}			-0.231 (0.157)
After First Bike Usage _{i,t} × $\log(\text{Bike Placement})_{i,t}$			0.070*** (0.013)
Individual FE	YES	YES	YES
Year-Month FE	YES	YES	-
City × Year-Month FE	NO	NO	YES
Clustered by City and Year-Month	YES	YES	YES
Sample	Full Sample	Full Sample	Bike Users
Observations	1,238,309	1,238,309	435,872
Adjusted R ²	0.800	0.800	0.774

Note: ***p < 0.01; **p < 0.05; *p < 0.01

10.3 Table 3: Bike usage intensity and heterogeneous placement effects

Interpretation. Placement effects are small for non-bike users and larger for one-time and repeat users, consistent with a mechanism operating through actual bike adoption/usage.

10.4 Table 4: In-person payment flow → credit provision (core 2SLS)

Interpretation. Instrumented payment flow increases both credit access and credit limits substantially; IV > OLS, consistent with endogeneity in payment/credit co-movement.

Table 3: Bike-usage Intensity and Heterogeneous Bike-placement Effects

This table reports the heterogeneous effects of city-level placement of shared bikes on the individual-level in-person payment flow and digital credit for non-bike users, one-time bike users, and repeat bike users. $\log(\text{Bike Placement})_{i,t}$ is the log number of active shared bikes in city c at time t . One-Time Bike User _{i,t} and Repeat Bike User _{i,t} are dummies for using a shared bike exactly once and at least twice from May 2017 to September 2020. $\log(1 + \text{In-Person Payment Flow})_{i,t}$ and $\log(1 + \text{Credit Line})_{i,t}$ are the $\log(1 + x)$ transformed amounts of in-person payment flow and credit line through Alipay in CNY. Columns (1) and (3) use individual and year-month fixed effects; columns (2) and (4) add city times year-month fixed effects. All standard errors are clustered at city and year-month level. ***, **, and * denote the 1%, 5%, and 10% confidence level, respectively. I report standard errors in parentheses.

	$\log(1 + \text{In-Person Payment Flow})_{i,t}$	$\log(1 + \text{Credit Line})_{i,t}$
	(1) (2) (3) (4)	
$\log(\text{Bike Placement})_{i,t}$	0.011 (0.009)	0.009 (0.010)
One-Time Bike User _{i,t} × $\log(\text{Bike Placement})_{i,t}$	0.086*** (0.020)	0.072*** (0.019)
Repeat Bike User _{i,t} × $\log(\text{Bike Placement})_{i,t}$	0.116*** (0.018)	0.078*** (0.017)
Individual FE	YES	YES
Year-Month FE	YES	-
City × Year-Month FE	NO	YES
Clustered by City and Year-Month	YES	YES
Observations	1,238,309	1,238,309
Adjusted R ²	0.352	0.555

Note: ***p < 0.01; **p < 0.05; *p < 0.1

Table 4: In-person Payment Flow and Credit Provision

The table shows the robust relationship between in-person payment flow (IPF) and BigTech credit with different specifications on both margins. Credit Access _{i,t} is a dummy for having access to Alipay's virtual credit card at time t . $\log(\text{Credit Line})_{i,t}$ is the log credit line conditional on Credit Access _{i,t} . Measure of In-Person Payment Flow _{i,t} is the amount of in-person payment flow through Alipay at time t , defined differently in different columns: $\log(1 + \text{In-Person Payment Flow})_{i,t}$ in CNY in columns (1) and (4); a dummy for positive In-Person Payment Flow _{i,t} in columns (2) and (3); $\log(\text{In-Person Payment Flow})_{i,t}$ conditional on positive In-Person Payment Flow _{i,t} in columns (3) and (6). $\log(\text{Bike Placement})_{i,t}$ is the log number of active shared bikes in city c at time t , conditional on positive $\log(\text{Bike Placement})_{i,t}$. The AR Statistic measures instrument strength using the Anderson-Rubin test, while its p-value provides additional evidence on instrument validity. They complement the conventional F-statistic. The F_{AR} Statistic represents the sample correlation between the main equation and first-stage residuals while imposing the null hypothesis. The VIF significance level is calculated by comparing the t-statistic against t-ratio critical values obtained using the VIF procedure (Lee et al., 2023). Panel A reports 2SLS estimates, instrumenting for payment flow with bike placement; Panel B reports the first stage; Panel C reports OLS estimates. All columns use individual and year-month fixed effects. All standard errors are clustered at city and year-month level. ***, **, and * denote the 1%, 5%, and 10% confidence level, respectively. I report standard errors in parentheses.

Panel A. Two-Stage Least Squares

	(1)	(2)	(3)	(4)	(5)	(6)
Measure of In-Person Payment Flow _{i,t}	0.016*** (0.024)	0.569*** (0.175)	0.087*** (0.083)	0.281*** (0.760)	2.033*** (0.011)	0.409*** (0.002)
$\log(\text{Bike Placement})_{i,t}$	0.011*** (0.010)	0.026*** (0.002)	0.039*** (0.012)	0.008*** (0.002)	0.008*** (0.014)	0.024*** (0.009)
F-Statistic	15.5	10.8	11.2	13.9	10.6	9.1
AR Statistic	86.9	36.9	12.4	35.1	35.1	18.4
AR Test P-Value	0.000	0.000	0.000	0.000	0.000	0.000
F_{AR} Statistic	0.091	0.093	0.049	0.078	0.062	0.079
VIF Significance Level	***	***	***	***	***	***
Adjusted R ²	0.351	0.465	0.432	0.327	0.439	0.401

Panel C. Ordinary Least Squares

	(1)	(2)	(3)	(4)	(5)	(6)
Measure of In-Person Payment Flow _{i,t}	0.016*** (0.001)	0.027*** (0.007)	0.007*** (0.001)	0.022*** (0.003)	0.072*** (0.023)	0.029*** (0.002)
Adjusted R ²	0.740	0.741	0.730	0.636	0.805	0.841
Form of the IPF Measure	$\log(1 + x)$	$1(x > 0)$	$\log(x)$	$\log(1 + x)$	$1(x > 0)$	$\log(x)$
Individual FE	YES	YES	YES	YES	YES	YES
Year-Month FE	YES	YES	YES	YES	YES	YES
Clustered by City and Year-Month	YES	YES	YES	YES	YES	YES
Sample	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample
Observations	1,238,309	1,238,309	662,010	779,283	779,283	516,570

Note: ***p < 0.01; **p < 0.05; *p < 0.01

Table 5: In-person Payment Flow and Consumer Behavior

This table presents empirical evidence showing the causal relationship between a user's in-person payment flow and the structure of the payment flows, in both the in-person payment and the online payment settings. Virtual Credit Card Share_{*i,t*} is the share of Alipay payments using the virtual credit card by individual *i* at time *t*. Compulsive Spending Share_{*i,t*} is the share of Alipay payments on cigarettes, games, lotteries, or live streaming services by individual *i* at time *t*. $\log(1 + \text{In-Person Payment Flow})_{i,t}$ is the $\log(1 + x)$ transformed amount of in-person payment flow through Alipay in CNY. $\log(\text{Bike Placement})_{i,t}$ is the log number of active shared bikes in city *c* at time *t*. Panel A reports 2SLS estimates, instrumenting for payment flow with bike placement; Panel B reports the first stage; Panel C reports OLS estimates. All columns use individual and year-month fixed effects. All standard errors are clustered at city and year-month level. ***, **, and * denote the 1%, 5%, and 10% confidence level, respectively. I report standard errors in parentheses.

	Virtual Credit Card Share _{<i>i,t</i>}		Compulsive Spending Share _{<i>i,t</i>}	
	In-Person Payment	Online Payment	In-Person Payment	Online Payment
	(1)	(2)	(3)	(4)
Panel A. Two-Stage Least Squares				
$\log(1 + \text{In-Person Payment Flow})_{i,t}$	0.014*** (0.004)	0.010*** (0.011)	0.004 (0.010)	0.002 (0.002)
Panel B. First Stage for $\log(1 + \text{In-Person Payment Flow})_{i,t}$				
$\log(\text{Bike Placement})_{i,t}$	0.026*** (0.009)	0.044*** (0.014)	0.028*** (0.009)	0.064*** (0.014)
F-Statistic	11.0	22.7	11.0	22.7
Adjusted R ²	0.434	0.505	0.434	0.505
Panel C. Ordinary Least Squares				
$\log(1 + \text{In-Person Payment Flow})_{i,t}$	0.010*** (0.002)	0.010*** (0.002)	0.003*** (0.003)	0.003*** (0.003)
Adjusted R ²	0.472	0.480	0.216	0.222
Individual FE	YES	YES	YES	YES
Year-Month FE	YES	YES	YES	YES
Clustered by City and Year-Month	YES	YES	YES	YES
Observations	662,010	806,938	662,010	806,938

Note: ***p < 0.01; **p < 0.05; *p < 0.1

10.5 Table 5: Payment flow and consumer behavior

Interpretation. Payment flow increases virtual-credit-card usage shares but does not increase compulsive spending shares in the 2SLS specification.

10.6 Table 6: Noncredit payment flow and credit provision

Interpretation. Noncredit payment flow has strong positive causal effects on credit access and credit limits, supporting the information channel.

Table 6: In-person Noncredit Payment Flow and Credit Provision

The table shows the causal relationship between in-person noncredit payment flow and BigTech credit on both margins. Credit Access_{*it*} is a dummy for having access to Alipay's virtual credit card at time *t*. $\log(\text{Credit Line})_{it}$ is the log credit line conditional on Credit Access_{*it*}. $\log(1 + \text{In-Person Noncredit Payment Flow})_{it}$ and $\log(1 + \text{In-Person Credit Payment Flow})_{it}$ are the $\log(1 + x)$ transformed amounts of in-person Alipay payment flow without and with the virtual credit card in CNY. $\log(\text{Bike Placement})_{it}$ is the log number of active shared bikes in city *c* at time *t*. Panel A reports 2SLS estimates, instrumenting for noncredit payment flow with bike placement; Panel B reports the first stage; Panel C reports OLS estimates. All columns use individual and year-month fixed effects. All standard errors are clustered at city and year-month level. ***, **, and * denote the 1%, 5%, and 10% confidence level, respectively. I report standard errors in parentheses.

	Credit Access _{<i>it</i>}		$\log(\text{Credit Line})_{it}$	
	(1)	(2)	(3)	(4)
Panel A. Two-Stage Least Squares				
$\log(1 + \text{In-Person Noncredit Payment Flow})_{it}$	0.094*** (0.024)	0.095** (0.026)	0.329*** (0.103)	0.358*** (0.124)
$\log(1 + \text{In-Person Credit Payment Flow})_{it}$		-0.005 (0.006)		-0.044 (0.029)
Panel B. First Stage for $\log(1 + \text{In-Person Noncredit Payment Flow})_{it}$				
$\log(\text{Bike Placement})_{it}$	0.037*** (0.009)	0.035*** (0.009)	0.037*** (0.010)	0.031*** (0.009)
$\log(1 + \text{In-Person Credit Payment Flow})_{it}$		0.218*** (0.009)		0.230*** (0.007)
F-Statistic	16.1	16.1	13.4	12.6
Adjusted R ²	0.475	0.492	0.457	0.480
Panel C. Ordinary Least Squares				
$\log(1 + \text{In-Person Noncredit Payment Flow})_{it}$	0.006*** (0.001)	0.004*** (0.001)	0.003 (0.002)	-0.004* (0.002)
$\log(1 + \text{In-Person Credit Payment Flow})_{it}$		0.015*** (0.001)		0.039*** (0.003)
Adjusted R ²	0.739	0.742	0.835	0.837
Individual FE	YES	YES	YES	YES
Year-Month FE	YES	YES	YES	YES
Clustered by City and Year-Month	YES	YES	YES	YES
Sample	Full Sample	Full Sample	Has Credit	Has Credit
Observations	1,238,309	1,238,309	779,283	779,283

Note: **p* < 0.1; ***p* < 0.05; ****p* < 0.01

Table 7: In-person Payment Flow and Credit Provision, with Enforcement Controls

The table shows the causal relationship between in-person payment flow and BigTech credit after controlling for time-varying assets under management (AUM) on both margins. Credit Access_{*it*} is a dummy for having access to Alipay's virtual credit card at time *t*. $\log(\text{Credit Line})_{it}$ is the log credit line conditional on Credit Access_{*it*}. $\log(1 + \text{In-Person Payment Flow})_{it}$ and $\log(1 + \text{Assets under Management})_{it}$ are the $\log(1 + x)$ transformed amounts of in-person payment flow and AUM on Alipay's platform in CNY. $\log(\text{Bike Placement})_{it}$ is the log number of active shared bikes in city *c* at time *t*. Panel A reports 2SLS estimates, instrumenting for payment flow with bike placement; Panel B reports the first stage; Panel C reports OLS estimates. All columns use individual and year-month fixed effects. All standard errors are clustered at city and year-month level. ***, **, and * denote the 1%, 5%, and 10% confidence level, respectively. I report standard errors in parentheses.

	Credit Access _{<i>it</i>}		$\log(\text{Credit Line})_{it}$	
	(1)	(2)	(3)	(4)
Panel A. Two-Stage Least Squares				
$\log(1 + \text{In-Person Payment Flow})_{it}$	0.097*** (0.025)	0.098*** (0.026)	0.280*** (0.085)	0.282*** (0.086)
$\log(1 + \text{Assets under Management})_{it}$	-0.005 (0.004)	-0.008 (0.005)	-0.015 (0.011)	-0.026* (0.013)
Panel B. First Stage for $\log(1 + \text{In-Person Payment Flow})_{it}$				
$\log(\text{Bike Placement})_{it}$	0.038*** (0.010)	0.036*** (0.010)	0.043*** (0.011)	0.043*** (0.011)
$\log(1 + \text{Assets under Management})_{it}$	0.147*** (0.005)	0.180*** (0.005)	0.122*** (0.005)	0.152*** (0.005)
F-Statistic	14.7	14.6	14.4	14.6
Adjusted R ²	0.562	0.566	0.533	0.536
Panel C. Ordinary Least Squares				
$\log(1 + \text{In-Person Payment Flow})_{it}$	0.009*** (0.001)	0.008*** (0.001)	0.020*** (0.002)	0.020*** (0.002)
$\log(1 + \text{Assets under Management})_{it}$	0.008*** (0.001)	0.008*** (0.001)	0.017*** (0.002)	0.014*** (0.002)
Adjusted R ²	0.741	0.742	0.836	0.836
Whether AUM Include Account Balance	NO	YES	NO	YES
Individual FE	YES	YES	YES	YES
Year-Month FE	YES	YES	YES	YES
Clustered by City and Year-Month	YES	YES	YES	YES
Sample	Full Sample	Full Sample	Has Credit	Has Credit
Observations	1,220,618	1,220,618	779,283	779,283

Note: **p* < 0.1; ***p* < 0.05; ****p* < 0.01

10.7 Table 7: Enforcement controls (AUM)

Interpretation. Controlling for AUM does not attenuate the payment-flow IV effect, suggesting enforcement is not driving the baseline relationship.

10.8 Table 8: Underserved segments

Interpretation. Low-education and older users have lower traditional finance engagement and lower “financial literacy” proxies, motivating heterogeneity analysis.

10.9 Table 9: Education heterogeneity

Interpretation. The causal effect is concentrated among the less educated; estimates for the more educated are small and often insignificant in 2SLS.

Table 8: Financially Underserved Segments

This table provides evidence that the less educated and the older tend to be financially underserved in China. Low Education, and Older than Median, are dummies for having no bachelor's degree or above and being older than half of the users. # Debit Cards_{*i*} is the number of debit cards linked to Alipay in April 2021. $\log(1 + \text{Max. AUM})_i$ is the log highest amount of AUM on Alipay from May 2017 to September 2020. # Investment Months_{*i*} is the months since first using Alipay's wealth management service till April 2021. Pay with Real Name_{*i*}, Use Own Account_{*i*}, and Complete Profile, are dummies for passing real name verification, using own account, and completing profile information in Alipay as of April 2021. Regression results show that the less educated and the older have lower financial service usage and literacy. All columns use city and gender fixed effects. All standard errors are clustered at city and year-month level. ***, **, and * denote the 1%, 5%, and 10% confidence level, respectively. I report standard errors in parentheses.

	Financial Service Usage			Financial Literacy		
	# Debit Cards _{<i>i</i>}	$\log(1 + \text{Max. AUM})_i$	# Investment Months _{<i>i</i>}	Pay with Real Name _{<i>i</i>}	Use Own Account _{<i>i</i>}	Complete Profile _{<i>i</i>}
	(1)	(2)	(3)	(4)	(5)	(6)
Low Education	-0.694*** (0.046)	-1.078*** (0.075)	-3.076*** (0.282)	-0.119*** (0.006)	-0.082*** (0.008)	-0.122*** (0.008)
Older than Median	-0.863*** (0.023)	-0.671*** (0.045)	-2.512*** (0.141)	-0.191*** (0.006)	-0.223*** (0.009)	-0.089*** (0.005)
Gender FE	YES	YES	YES	YES	YES	YES
City FE	YES	YES	YES	YES	YES	YES
Year-Month FE	YES	YES	YES	YES	YES	YES
Clustered by City	YES	YES	YES	YES	YES	YES
Observations	39,459	39,459	39,459	39,459	39,459	39,459
Adjusted R ²	0.081	0.052	0.036	0.081	0.101	0.046

Note:

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Table 9: Education, In-person Payment Flow, and Credit Provision

The table shows the causal relationship between in-person payment flow and BigTech credit for the less and more educated groups on both margins. Credit Access_{*i,t*} is a dummy for having access to Alipay's virtual credit card at time *t*. $\log(\text{Credit Line})_{i,t}$ is the log credit line conditional on Credit Access_{*i,t*}. $\log(1 + \text{In-Person Payment Flow})_{i,t}$ is the $\log(1 + x)$ transformed amount of in-person payment flow through Alipay in CNY at time *t*. $\log(\text{Bike Placement})_{i,t}$ is the log number of active shared bikes in city *c* at time *t*. Panel A reports 2SLS estimates, instrumenting for payment flow with bike placement; Panel B reports the first stage; Panel C reports OLS estimates. All columns use individual and year-month fixed effects. Columns (1) and (3) use the less educated subsample with no college degree or above; columns (2) and (4) use the more educated subsample with a bachelor's degree or above. All standard errors are clustered at city and year-month level. ***, **, and * denote the 1%, 5%, and 10% confidence level, respectively. I report standard errors in parentheses.

	Credit Access _{<i>i,t</i>}		$\log(\text{Credit Line})_{i,t}$	
	(1)	(2)	(3)	(4)
Panel A. Two-Stage Least Squares				
$\log(1 + \text{In-Person Payment Flow})_{i,t}$	0.093*** (0.027)	0.024 (0.044)	0.334*** (0.109)	0.038 (0.073)
Panel B. First Stage for $\log(1 + \text{In-Person Payment Flow})_{i,t}$				
$\log(\text{Bike Placement})_{i,t}$	0.039*** (0.010)	0.043*** (0.013)	0.039*** (0.011)	0.053*** (0.014)
F-Statistic	13.7	10.9	11.6	14.2
Adjusted R ²	0.554	0.563	0.528	0.483
Panel C. Ordinary Least Squares				
$\log(1 + \text{In-Person Payment Flow})_{i,t}$	0.009*** (0.001)	0.013*** (0.001)	0.022*** (0.003)	0.013*** (0.002)
Adjusted R ²	0.741	0.734	0.831	0.893
Individual FE	YES	YES	YES	YES
Year-Month FE	YES	YES	YES	YES
Clustered by City and Year-Month	YES	YES	YES	YES
Sample	Full Sample	Full Sample	Has Credit	Has Credit
Subsample	Low Education	High Education	Low Education	High Education
Observations	1,065,769	171,938	657,878	121,194

Note:

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

11 Conclusion

11.1 Summary: what the paper concludes

This paper provides causal evidence that adopting cashless payment can *expand BigTech consumer credit supply* and improve *financial inclusion*. The core empirical challenge is endogeneity: payment usage and credit outcomes co-move because (i) more creditworthy users may adopt cashless payment earlier, and (ii) credit availability can itself increase payment activity.

The paper addresses this by exploiting quasi-exogenous variation in cashless payment adoption generated by the staggered city-level rollout of *Alipay-bundled dockless shared bikes*. Bike placement induces users to set up and use Alipay for in-person transactions, which in turn generates payment-flow data that the platform can use for screening and credit-limit setting.

The main conclusions are:

- (i) **In-person payment flow causally increases credit access (extensive margin).** In the baseline 2SLS specifications (with individual and year-month fixed effects), higher in-person payment activity increases the probability that a user has access to Alipay's virtual credit card. For example, instrumenting in-person payment flow with bike placement, the estimated effect is about 0.086 on *CreditAccess* when the regressor is $\log(1 + IPF)$, and about 0.563 when the regressor is the indicator $\mathbf{1}\{IPF > 0\}$ (Table 4, Panel A).
- (ii) **In-person payment flow causally increases credit limits (intensive margin).** Conditional on having access, payment flow strongly increases credit limits. For example, the effect is about 0.281 on $\log(\text{CreditLine})$ when the regressor is $\log(1 + IPF)$, about 2.033 when the regressor is $\mathbf{1}\{IPF > 0\}$, and about 0.409 when the regressor is $\log(IPF)$ conditional on $IPF > 0$ (Table 4, Panel A).

- (iii) **The mechanism is primarily an information channel (transaction data), not merely enforcement/collateral.** When decomposing payment into *noncredit* (not paid by the virtual credit card) and *credit-paid* components, the causal effect is driven by **noncredit** in-person payment flow: $\log(1 + \text{Noncredit IPF})$ strongly predicts both access and limits, while the credit-paid component is small (Table 6). Moreover, controlling for assets under management (AUM) on the platform does not attenuate the payment-flow effect, suggesting that the main results are not just “implicit collateral” (Table 7).
- (iv) **Cashless payment is associated with financial inclusion: the underserved benefit more.** The causal effect is substantially larger for users who are plausibly underserved by traditional finance. For instance, using $\log(1 + \text{IPF})$, the 2SLS effect is large and significant for the *less educated* (e.g., 0.093 on access and 0.334 on log limits) and small/insignificant for the more educated (Table 9). Parallel results hold for older users in the appendix.
- (v) **There is no evidence (in the paper’s measured proxy) that the inclusion channel works through more “compulsive” spending.** Instrumented payment flow increases the share of payments made using the virtual credit card (consistent with expanded credit), but has no statistically robust effect on the share of spending classified as compulsive (Table 5).

11.2 Economic intuition: why cashless payment can expand credit and inclusion

The paper’s intuition is that cashless payment adoption creates a *new information infrastructure*:

1) Cashless payment generates verifiable, high-frequency data. In-person transaction flows reveal patterns that are hard for traditional lenders to observe at scale: regularity of spending, stability over time, category composition, and indicators of local economic activity.

2) Data reduces screening frictions and expands supply. When borrower quality is imperfectly observed, lenders either ration credit or set conservative limits. Richer transaction data improves inference about borrower types and repayment capacity, so the platform can:

- grant access to users who previously looked “opaque” under traditional data,
- increase limits where the data indicate higher capacity,
- and tighten limits for signals of risk (a point emphasized by the paper’s illustrative theory in Figure 6).

3) Why the shared-bike instrument matters. Shared bikes act as a shock that lowers setup and learning costs of using Alipay for offline payments. Once users start paying via Alipay, the platform observes broader non-bike payment behavior, which is exactly the input into its credit supply rule. Thus the identification chain is:

Bike placement $\rightarrow$ cashless-payment adoption / higher *IPF* $\rightarrow$ expanded credit access and higher limits.

4) **Why inclusion effects are stronger for underserved users.** Underserved users often have weaker traditional documentation (thin credit files, fewer bank relationships). For them, transaction data is a closer substitute for missing formal financial records, so the marginal informational gain is larger, leading to larger expansions in access and limits.

11.3 Scope and interpretation (what the paper does and does not claim)

- The IV estimates identify a **local causal effect** for “compliers” whose payment behavior responds to bike placement. The paper is not claiming that *all* payment usage mechanically causes credit in all contexts; the design isolates an exogenous adoption margin.
- The outcomes are unusually supply-like because the platform reveals access and limits without a standard application process, but a full welfare analysis (defaults, debt burdens, consumption smoothing) requires additional data beyond what is emphasized here.
- External validity likely depends on institutional features (data privacy regimes, credit reporting, platform market power), but the mechanism—*data-driven screening expanding credit*—is portable to other cashless-payment environments.

11.4 Takeaway

By using shared-bike rollout as an instrument for cashless-payment adoption, Ouyang shows that in-person payment flow causally expands BigTech consumer credit access and credit limits, with especially large effects for financially underserved users, consistent with transaction data alleviating information frictions and enabling financial inclusion.